Functional

The user can move around in the open-world map. They can interact with NPCs, purchase items from shops, and embark on an adventure. The user may encounter enemies on their journey. When encountering an enemy, a battle simulation plays. In the battle simulation, the user is given the choice of what to do for each member of their team, whether that be to attack, defend, use items, or run. If the user wins this encounter, they are rewarded and are able to move forward. However, if the user loses the encounter, they lose all their progress and must go back.

Non-functional

The interface and user inputs should load instantly.

Constraints and Assumptions

Since Python isn't as strong as C# or C++ for OOP (object orientated programming), it may have trouble handling multiple classes at once within a while loop (multithreading).

Application Architecture

During the battle simulation, the user is prompted with various actions. Based on their actions, it changes the outcome for the encounter. For instance, if the user chooses to attack an enemy, the program will process that action, reducing the health of that enemy and displaying the action through the interface.

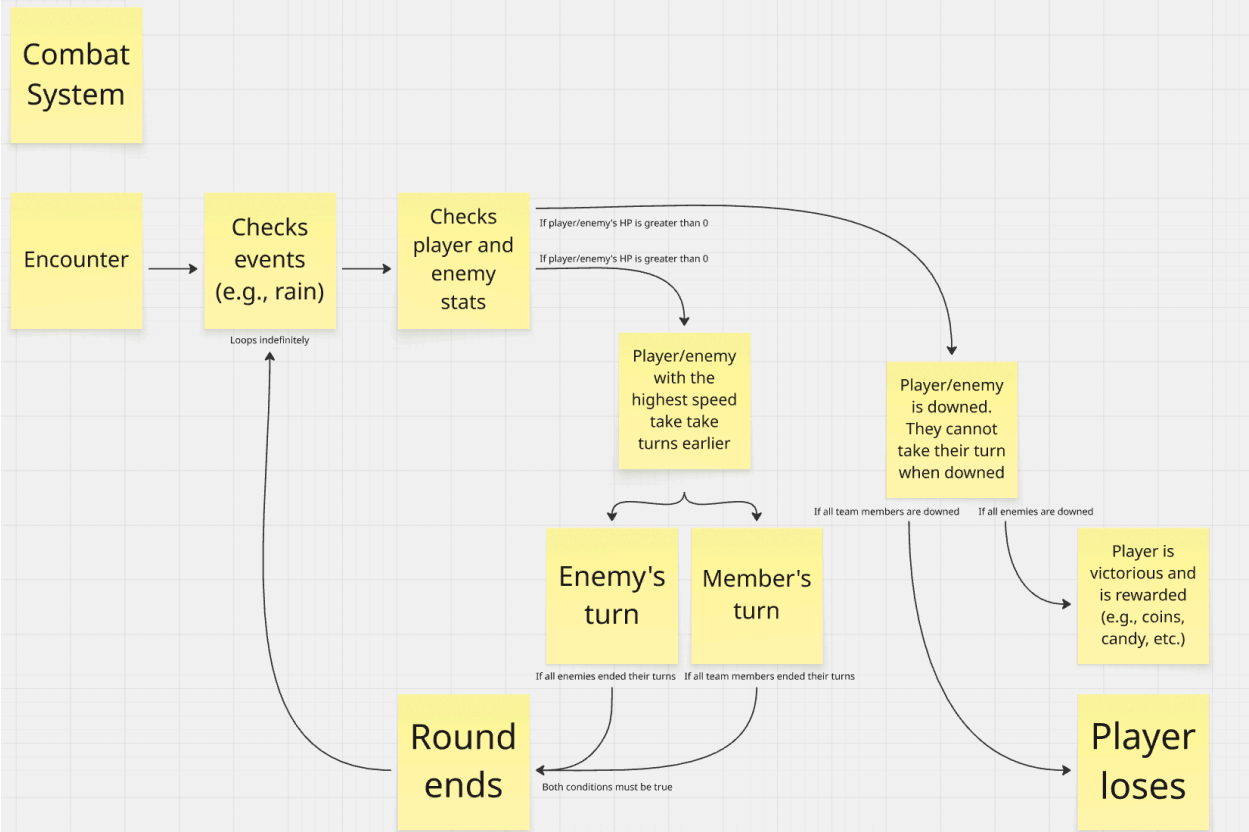
Key Algorithms

- Display the stats for both team members enemies stats during encounters
- Prompt the user an action for each team member
- Reward the user after succeeding an encounter
- Prompt the user if they'd to purchase something
- User cannot afford items if they do not have enough money or inventory space

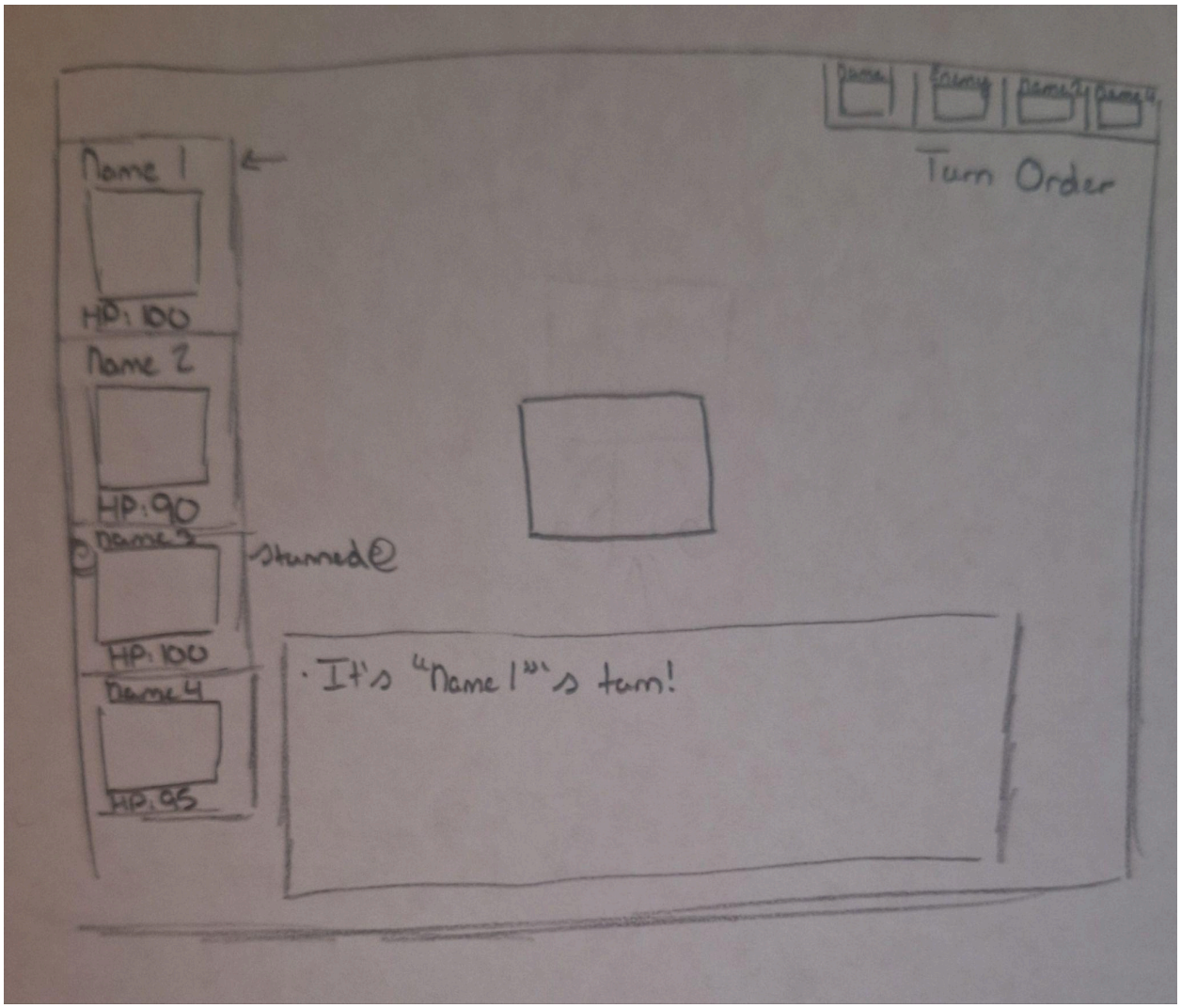
Primary Data Structures

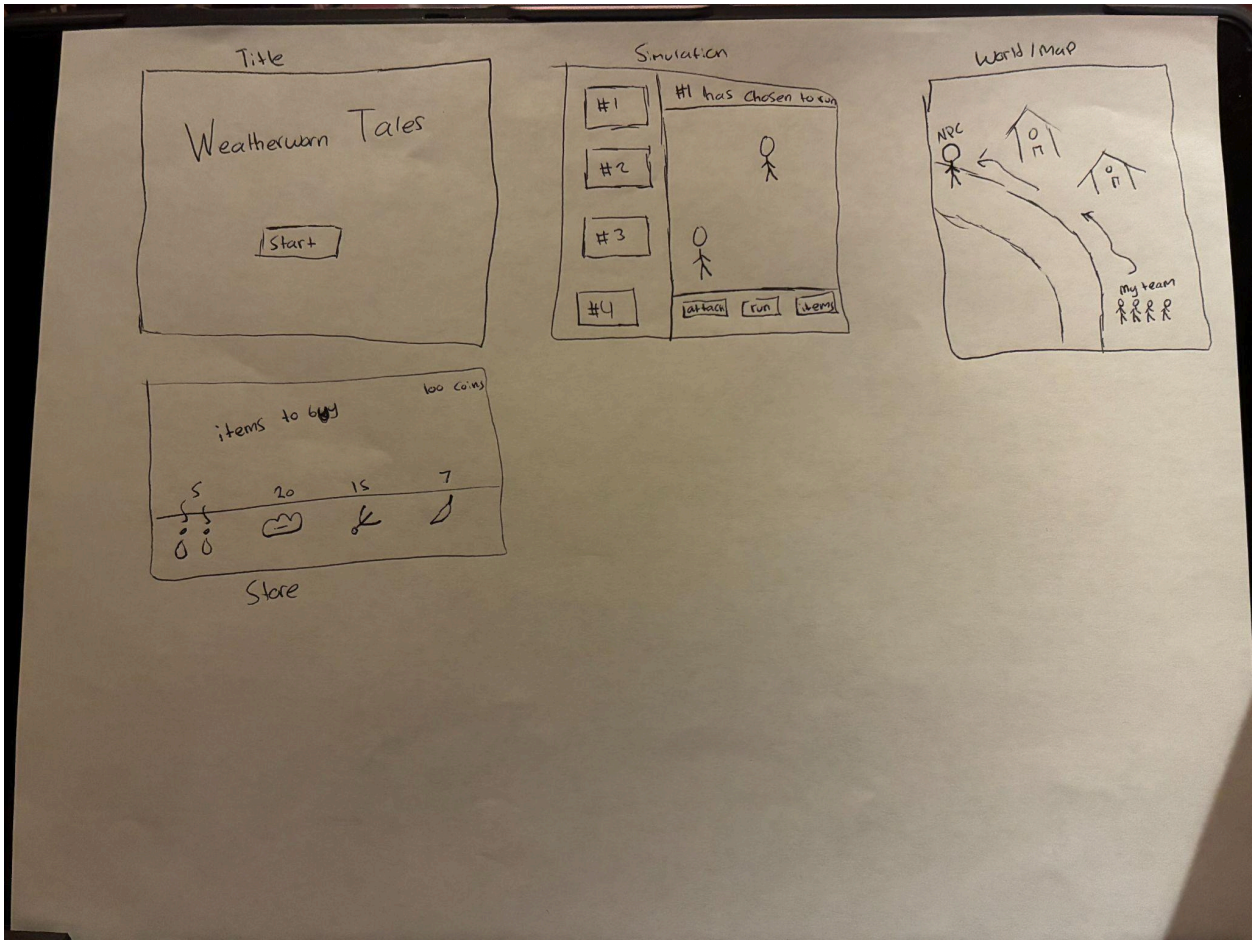
- Objects/Classes
- Lists
- Loops
- Hash Maps

Flowchart



User Interface Design




Title
Weatherworn Tales
Start
Simulation
#1
#2
#3
#4
#1 has chosen to run
attack
run
items
World / Map
NPC
my team
items to buy
100 coins
5
20
15
7
Store

GitHub Task Log & Project Management Document

- Include a screen or export of your group's GitHub Issues or GitHub Projects Board showing the breakdown of tasks for Phase 2
- Clearly document which team member was assigned to each Phase 2 requirement (Requirements Writing, System Architecture, ERD, Pseudocode, Wireframes)
- Provide the active link to your team's official GitHub repository with confirmed access for course instructors
    - https://github.com/magj09/cs-sdlc-pwingu